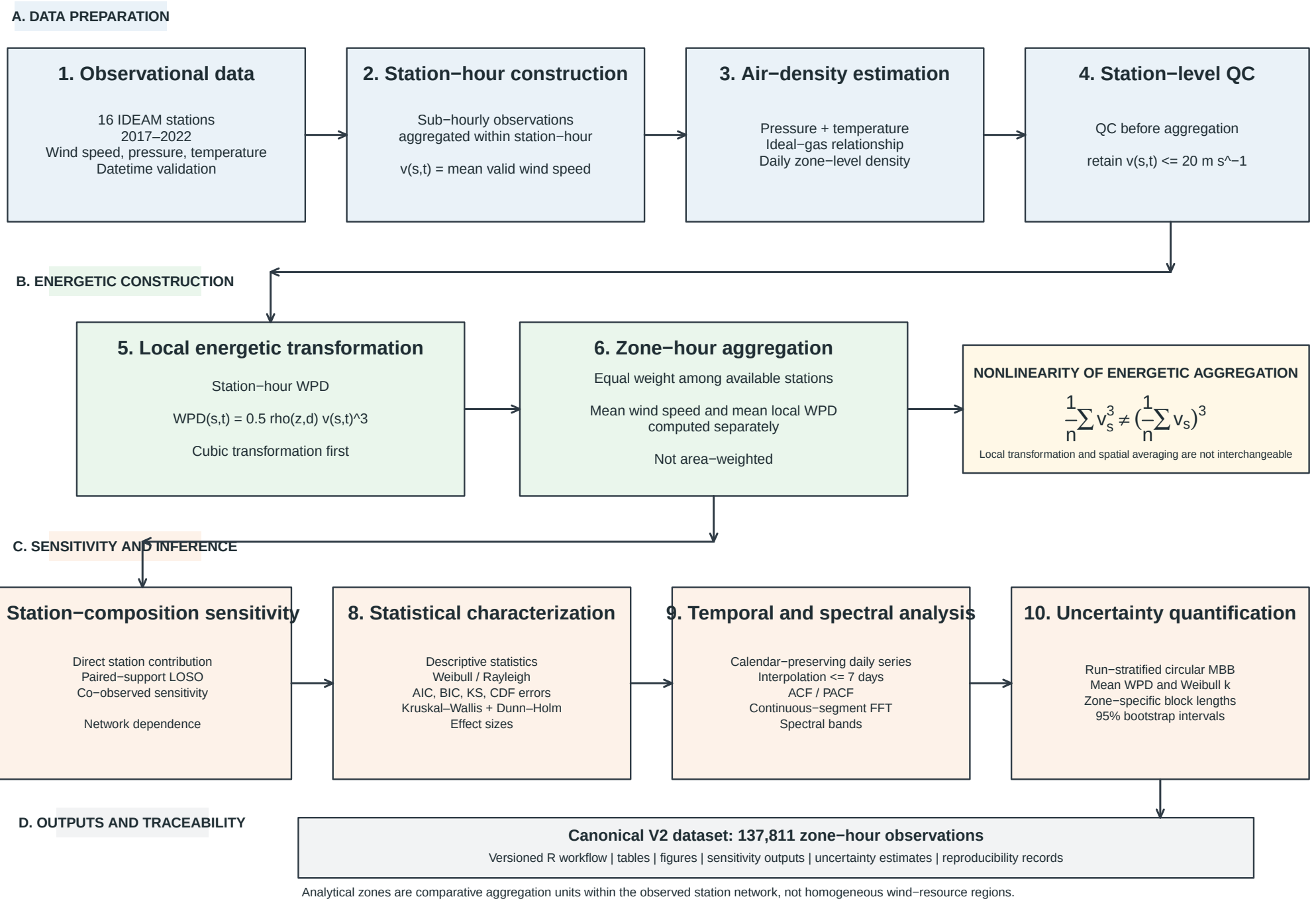
A. DATA PREPARATION
1. Observational data
16 IDEAM stations
2017–2022
Wind speed, pressure, temperature
Datetime validation
2. Station−hour construction
Sub−hourly observations
aggregated within station−hour
v(s,t) = mean valid wind speed
3. Air−density estimation
Pressure + temperature
Ideal−gas relationship
Daily zone−level density
4. Station−level QC
QC before aggregation
retain v(s,t) <= 20 m s^−1
B. ENERGETIC CONSTRUCTION
5. Local energetic transformation
Station−hour WPD
WPD(s,t) = 0.5 rho(z,d) v(s,t)^3
Cubic transformation first
6. Zone−hour aggregation
Equal weight among available stations
Mean wind speed and mean local WPD
computed separately
Not area−weighted
NONLINEARITY OF ENERGETIC AGGREGATION
$\frac{1}{n}\sum v_s^3 \neq (\frac{1}{n}\sum v_s)^3$
Local transformation and spatial averaging are not interchangeable
C. SENSITIVITY AND INFERENCE
7. Station−composition sensitivity
Direct station contribution
Paired−support LOSO
Co−observed sensitivity
Network dependence
8. Statistical characterization
Descriptive statistics
Weibull / Rayleigh
AIC, BIC, KS, CDF errors
Kruskal–Wallis + Dunn–Holm
Effect sizes
9. Temporal and spectral analysis
Calendar−preserving daily series
Interpolation <= 7 days
ACF / PACF
Continuous−segment FFT
Spectral bands
10. Uncertainty quantification
Run−stratified circular MBB
Mean WPD and Weibull k
Zone−specific block lengths
95% bootstrap intervals
D. OUTPUTS AND TRACEABILITY
Canonical V2 dataset: 137,811 zone−hour observations
Versioned R workflow | tables | figures | sensitivity outputs | uncertainty estimates | reproducibility records
Analytical zones are comparative aggregation units within the observed station network, not homogeneous wind−resource regions.